

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Document Enterprise Cybersecurity Issues

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Objectives

- Part 1: Record your assessment of Athena's cybersecurity issues.
- Part 2: Record the different types of assets owned by Athena.
- Part 3: List the threats for each asset type.
- Part 4: Recommend mitigation techniques to address each threat.

Scenario Summary

Athena Learning Incorporated is an educational service provider with two core lines of business: course content creation and online learning services. Athena employs approximately 100 staff at its headquarters and 5 staff each in London and Singapore. The company manages student accounts, payment information, and intellectual property belonging to both Athena and its partners. Athena operates a mixed Windows/Apple/Linux environment, uses cloud services for course delivery, allows BYOD, supports remote workers, and self-hosts DNS, email, and intranet services.

Part 1: Assessment of Athena's Cybersecurity Issues

After reviewing the Athena scenario, the following key cybersecurity issues and risk areas were identified:

BYOD (Bring Your Own Device): Employees use personal phones and tablets for work. These devices are not fully managed by IT, introducing risks of data leakage, malware infection, and unauthorized data access.

Remote Access: Some employees require full network access from home. VPN and remote access introduce additional attack surfaces if not properly secured and monitored.

Mixed and Outdated Operating Systems: The network uses various PC brands, models, and different versions of operating systems. Unpatched or unsupported OS versions are highly vulnerable to exploitation.

Partner Access via Web Portal: External partners have access to parts of the internal network through a web portal. This third-party access must be tightly controlled to prevent unauthorized data access or injection attacks.

SQL Databases Connected to Web Portals: SQL databases accessible through web portals are vulnerable to SQL injection attacks, which could expose or corrupt student records and payment information.

Student Payment and Personal Data: Athena manages student registration, authentication, records, and payment information — all highly sensitive data requiring strong encryption and access controls.

Cloud Services with Internal Mirroring: Course content is assembled internally and then mirrored to cloud services. Data in transit between internal systems and the cloud must be encrypted and protected.

Self-Hosted DNS, Email, and Intranet: Operating these services internally increases the attack surface. DNS poisoning, email phishing, and intranet compromise are all significant risks.

Custom and Internally Developed Software: Custom applications may contain unvetted vulnerabilities that are not covered by standard security patches, making them harder to protect and maintain.

Multi-Location Operations: Operating across HQ, London, and Singapore introduces challenges in enforcing consistent security policies and standards across all regions.

Part 2: Assets Owned by Athena

The following tables categorize Athena's assets across four asset types.

Table 1: Information / Data Assets

Information / Data Asset	Description
Student Account Information	Student registration details, authentication credentials, learning records and progress data.
Student Payment Information	Credit/debit card details, billing information and transaction records for course purchases.
Course Content (Athena-owned)	Text, graphics, video and interactive learning assets created by Athena — core intellectual property.
Partner Course Content	Content belonging to Athena's partners, hosted and managed by Athena on their behalf.
Employee Records	HR data including personal details, contracts, payroll and employment history.
Financial and Business Records	Accounting data, invoices, contracts and internal business documents.
SQL Database Contents	Structured data stored in Athena's SQL databases, including student, partner and business data.

Table 2: Software Assets

Software Asset	Description
Microsoft Windows OS (various versions)	Operating system used on most client workstations across all offices.
Apple iOS / macOS	Operating system used on Apple client devices and personal BYOD devices.
Microsoft Server Software	Server operating system and services (Active Directory, file sharing, etc.) on Windows servers.
Linux Server OS	Used on Athena servers for hosting learning content assets, financial data and business records.
SQL Database Software	Database management system hosting student, partner and business data connected to web portals.

Learning Management System (LMS)	Platform used to assemble, manage and deliver course content to students.
Custom / Internally Developed Applications	Bespoke software built by Athena staff for specific business functions.
Office Application Software	Commercial productivity software such as Microsoft Office used by all employees.
Web Portal Software	The secure web portal that provides partner access to the internal network.
Cloud Services Platform	Cloud infrastructure used to host and deliver courses to students globally.

Table 3: Physical Assets

Physical Asset	Description
Employee Workstations / PCs	Various brands and models of computers used by staff at all office locations.
Servers (Windows and Linux)	Physical server hardware storing content, databases, employee records and financial data.
Personal Mobile Devices (BYOD)	Employee-owned smartphones and tablets used for work under the BYOD policy.
Network Hardware	Routers, switches, firewalls and access points forming the Athena network infrastructure.
Office Facilities	Physical buildings at headquarters, London and Singapore offices.
External Storage Media	USB drives, backup drives and other removable storage used by staff.

Table 4: Network Assets

Network Asset	Description
Internal Corporate Network	The main LAN connecting all workstations, servers and network devices at each office.
VPN (Virtual Private Network)	Secure tunnel enabling remote workers to access the internal network from home.
Cloud Services Infrastructure	Network connections to the cloud platform used to deliver courses publicly.
Partner Web Portal	Secure web-based interface providing controlled external access to parts of the network.
DNS Service (Self-hosted)	Athena's internal and external domain name resolution service.
Email Service (Self-hosted)	Athena's internal email server handling all corporate communications.
Intranet	Internal web-based platform for company-wide communication and information sharing.
WAN / Internet Connections	Internet connectivity at headquarters, London and Singapore offices.

Part 3: Threats for Each Asset Type

3a. Difference Between a Threat and a Vulnerability

A vulnerability is a weakness or flaw in a system, process, or configuration that could be exploited. A threat is the potential danger or actor that could exploit that vulnerability to cause harm. For example, an unpatched operating system is a vulnerability. A hacker who targets that unpatched OS with malware is the threat. Vulnerabilities exist independently; threats require a vulnerability to be effective. Together they contribute to risk: $\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Asset Value}$.

3b. Threats for Each Asset

Information / Data Assets — Threats

Information / Data Asset	Threats
Student Account & Payment Information	SQL Injection; Phishing attacks targeting credentials; Ransomware encrypting records; Insider theft; Man-in-the-Middle (MitM) interception; Unauthorized access due to weak passwords.
Course Content (Athena & Partner)	Intellectual property theft; Unauthorized copying or distribution; Ransomware; Insider threat by disgruntled employee; Data corruption.
Employee Records	Insider data theft; Unauthorized access; Data breach via compromised accounts; Social engineering targeting HR staff.
Financial and Business Records	Financial fraud; Ransomware; Unauthorized access; Data exfiltration by malicious insiders; Phishing targeting finance staff.
SQL Database Contents	SQL Injection attacks; Privilege escalation; Unauthorized database access; Data manipulation or deletion.

Software Assets — Threats

Software Asset	Threats
Windows / macOS / Linux OS	Exploitation of unpatched vulnerabilities; Malware and ransomware; Zero-day exploits; Privilege escalation attacks.
SQL Database Software	SQL Injection; Unauthorized administrative access; Exploitation of known database CVEs; Misconfiguration.
Custom / Internally Developed Applications	Unvetted code vulnerabilities (e.g., XSS, buffer overflow); Lack of security patching; Logic flaws exploited by attackers.
Web Portal Software	Cross-site scripting (XSS); Brute-force login attacks; Session hijacking; Unauthorized access by partners.
Cloud Services Platform	Account hijacking; Misconfigured cloud storage (data exposure); API vulnerabilities; Data interception in transit.

Physical Assets — Threats

Physical Asset	Threats
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Employee Workstations / PCs	Malware via USB or email; Physical theft; Unauthorized physical access; Keyloggers installed by unauthorized persons.
Servers	Physical theft or damage; Unauthorized physical access to server room; Natural disasters (fire, flood); Hardware failure.
Personal Mobile Devices (BYOD)	Device theft or loss exposing company data; Malware from personal app stores; Unencrypted company data on device; Unsecured Wi-Fi connections.
Network Hardware	Physical tampering; Unauthorized device connection; Theft of hardware; Power failure causing network outage.
External Storage Media	Data theft via USB drives; Malware introduction via infected media; Loss of unencrypted backup drives.

Network Assets — Threats

Network Asset	Threats
Internal Corporate Network	Unauthorized access; Lateral movement by attackers; Packet sniffing / eavesdropping; Rogue devices connecting to the network.
VPN	Brute-force attacks on VPN credentials; Exploitation of VPN vulnerabilities; Unauthorized remote access; Split tunneling misuse.
Cloud Services Infrastructure	DDoS attacks disrupting course delivery; Data interception during cloud sync; Misconfigured cloud permissions.
Partner Web Portal	Unauthorized partner access; Credential sharing; SQL Injection via portal; Malicious file upload by partner.
DNS Service	DNS poisoning / cache poisoning; DNS hijacking redirecting users to malicious sites; DDoS targeting DNS.
Email Service	Phishing and spear-phishing attacks; Malware distribution via email attachments; Email spoofing; Business Email Compromise (BEC).
Intranet	Unauthorized access to sensitive internal documents; Cross-site scripting (XSS); Malicious insider publishing false content.

Part 4: Mitigation Techniques for Each Threat

Information / Data Assets — Mitigation

Information / Data Asset	Threats	Mitigation Techniques
Student Account & Payment Info	SQL Injection; Phishing; Ransomware; MitM; Weak passwords	Input validation and parameterized queries; MFA; AES-256 encryption at rest and in transit (TLS); Regular backups; PCI-DSS compliance; Strong password policy.
Course Content (Athena & Partner)	IP theft; Ransomware; Insider threat	Role-based access control (RBAC); DRM for content; Offline backups;

		Watermarking of partner content; Audit logs on access.
Employee Records	Unauthorized access; Data breach; Social engineering	RBAC — HR access only; Encryption; Security awareness training; Regular access reviews.
Financial Records	Fraud; Ransomware; Phishing	MFA on finance systems; Segregation of duties; Encrypted backups; Email filtering; SOX controls.
SQL Database Contents	SQL Injection; Privilege escalation	Parameterized queries; Principle of least privilege on DB accounts; WAF (Web Application Firewall); DB activity monitoring.

Software Assets — Mitigation

Software Asset	Threats	Mitigation Techniques
Windows / macOS / Linux OS	Unpatched vulnerabilities; Malware; Zero-day	Regular patch management and automatic updates; Endpoint antivirus/EDR; Application whitelisting; OS hardening (disable unused services).
SQL Database Software	SQL Injection; Unauthorized access; Misconfiguration	Parameterized queries; DB firewall; Disable default accounts; Regular security audits; Encrypted connections.
Custom Applications	Code vulnerabilities; Logic flaws	Secure Software Development Lifecycle (SSDLC); Code reviews and static analysis; Penetration testing before deployment; Bug bounty program.
Web Portal Software	XSS; Brute-force; Session hijacking	Input sanitization; WAF; Account lockout policies; Secure session management; HTTPS/TLS.
Cloud Services Platform	Account hijacking; Misconfiguration; API vulnerabilities	MFA on cloud accounts; Cloud security posture management (CSPM); API gateway with authentication; Encryption in transit; Regular audits.

Physical Assets — Mitigation

Physical Asset	Threats	Mitigation Techniques
Employee Workstations	Malware; Theft; Unauthorized access	Full-disk encryption (BitLocker/FileVault); Cable locks; Screen lock policies; Antivirus/EDR; Restrict USB ports via Group Policy.
Servers	Physical theft; Unauthorized access; Disaster	Locked server room with access control; CCTV; UPS and redundant power; Offsite backups; Fire suppression systems.
Personal Mobile Devices (BYOD)	Device theft; Malware; Unencrypted data	MDM enrollment; Remote wipe capability; Mandatory encryption;

		Containerization of work data; Prohibited access to unsecured Wi-Fi.
Network Hardware	Tampering; Rogue devices; Power failure	Physical access controls; Port security (802.1X); UPS; Network monitoring tools; Disable unused ports.
External Storage Media	Data theft; Malware via USB	USB port restrictions via Group Policy; Encrypted USB drives only; Logging of media usage; Policy prohibiting personal USB devices.

Network Assets — Mitigation

Network Asset	Threats	Mitigation Techniques
Internal Corporate Network	Unauthorized access; Eavesdropping; Rogue devices	Network segmentation (VLANs); 802.1X port authentication; IDS/IPS; Encrypted internal traffic; Regular network scans.
VPN	Brute-force; Credential theft; Unauthorized access	MFA on VPN; Strong encryption (IKEv2/IPSec); Monitor VPN logs; Certificate-based authentication; Split tunneling controls.
Cloud Services Infrastructure	DDoS; Data interception; Misconfiguration	DDoS protection (CDN/scrubbing); TLS encryption; Cloud security audits; Strict IAM policies; Content integrity checks.
Partner Web Portal	Unauthorized access; SQL Injection; Malicious upload	Dedicated portal with minimal network access; WAF; File upload scanning; Partner account MFA; Regular access reviews.
DNS Service	DNS poisoning; Hijacking; DDoS	DNSSEC implementation; DNS monitoring and alerting; Redundant DNS servers; Rate limiting on DNS queries.
Email Service	Phishing; Malware; Spoofing; BEC	SPF, DKIM and DMARC records; Email filtering (anti-spam/anti-phishing); User security awareness training; Email sandboxing for attachments.
Intranet	Unauthorized access; XSS; Insider misuse	Authentication required for all access; Input sanitization; RBAC on intranet content; Audit logs on document access.

Reflection

1. Why is it useful to categorize assets when identifying threats and mitigation techniques?

Categorizing assets is useful because different asset types face fundamentally different threats and require different mitigation strategies. For example, physical assets are vulnerable to theft and physical damage, while information assets are vulnerable to data breaches and ransomware. By grouping assets into categories — data, software, physical, and network — security teams can apply targeted, appropriate controls rather than generic solutions. Categorization also helps prioritize resources: high-value assets such as student payment data receive the strongest protections. It ensures nothing is overlooked, makes assigning ownership and accountability easier, and provides a structured foundation for compliance with regulations such as GDPR and PCI-DSS.

2. Do some threats have the same or similar mitigation measures? Why is it important to note this?

Yes — many threats share common mitigation techniques. For example, multi-factor authentication (MFA) mitigates unauthorized access across email, VPN, cloud platforms, and the partner web portal. Encryption addresses both data interception and physical device theft. Regular patching reduces exposure to unpatched vulnerabilities across all software assets. Security awareness training helps defend against phishing, social engineering, and insider threats simultaneously. Recognizing these overlaps is important because it allows organizations to invest strategically — a single well-implemented control can reduce risk across multiple asset categories at once. It also supports the Defense-in-Depth strategy, where the same mitigation applied at multiple layers provides compounding protection.

3. What have you learned about applying cybersecurity knowledge to a real organization?

This lab demonstrated that real organizations have complex, interconnected assets that cannot be protected in isolation. A single vulnerability — such as an unpatched SQL database connected to a web portal — can expose student payment data, partner content, and business records simultaneously. It also highlighted that technical controls alone are insufficient: Athena's risks include BYOD policies, remote access, human error, and third-party partner access, all of which require procedural and policy-based controls in addition to technology. The lab reinforced the importance of a layered security approach, consistent policy enforcement across all locations, and continuous review as the threat landscape evolves. Understanding an organization's business context is essential to identifying what matters most and where defenses must be strongest.